

12-xalqaro fizika olimpiadasining masalalari

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Manba: O. F. Kabardin, V. A. Orlov, "International Physics Olympiads for High School Students", tahrir. V. G. Razumovskiy, Moskva, Nauka, 1985. (Rus tilida).

Nazariy masala 1

Massasi M va silindrsimon shakldagi qo'zg'almas idish vakuumda joylashtirilgan. Uning bir uchi yopiq. Massasi m va eni hisobga olmaslik darajada kichik bo'lgan qo'zg'almas porshen idish hajmini ikkita teng qismga ajratadi. Yopiq qismda molyar massasi M_0 va harorati T bo'lgan n mol bir atomli ideal gaz mavjud. Porshen qo'yib yuborilgach, u idishdan ishqalanishsiz chiqib ketadi. So'ngra gaz ham idishni tark etadi. Idishning oxirgi tezligi qanday bo'ladi?

Gaz doimiysi R . Porshen chiqib ketgunga qadar gazning impulsi hisobga olinmasa bo'ladi. Gaz, idish va porshen orasida issiqlik almashinuvi yo'q. Gaz idishni tark etayotganda uning haroratining o'zgarishi hisobga olinmasligi mumkin. Yerning tortishish kuchini hisobga olmang.

Nazariy masala 1 yechimi

Porshen idishni tark etgunga qadar tizimni yopiq tizim deb hisoblash mumkin. Impuls va energiyaning saqlanish qonunlaridan:

$$(M + nM_0)v_1 - mu = 0, \quad (1)$$

$$\frac{(M + nM_0)v_1^2}{2} + \frac{mu^2}{2} = \Delta U, \quad (2)$$

bu yerda v_1 – porshen idishni tark etayotgan paytdagi idish tezligi, u – porshenning shu paytdagi tezligi, ΔU – gaz ichki energiyasining o'zgarishi. Gaz ideal va bir atomli, shuning uchun:

$$\Delta U = \frac{3}{2}nR\Delta T = \frac{3}{2}nR(T - T_f); \quad (3)$$

T_f – porshen idishni tark etayotgan paytdagi gaz harorati. Bu harorat adiabatik jarayon qonuni orqali aniqlanadi:

$$pV^\gamma = \text{const.}$$

Ideal gaz tenglamasi $pV = nRT$ dan foydalanib,

$$TV^{\gamma-1} = \text{const.}, \quad TV^{\gamma-1} = T_f V_f^{\gamma-1}.$$

$V_f = 2V$ munosabatdan va bir atomli gaz uchun adiabat ko'rsatkichi $\gamma = \frac{c_p}{c_v} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \frac{5}{3}$ ekanligidan foydalanib, oxirgi harorat:

$$T_f = T \left(\frac{V}{V_f} \right)^{\gamma-1} = \frac{T}{2^{2/3}} = T \cdot 2^{-2/3}. \quad (4)$$

(1)–(4) tenglamalarni yechib, quyidagini olamiz:

$$v_1 = \sqrt{3(1 - 2^{-2/3}) \frac{mnRT}{(nM_0 + M)(m + nM_0 + M)}}. \quad (5)$$

Agar gaz massasi nM_0 idish massasi M va porshen massasi m dan ancha kichik bo'lsa, (5) tenglama soddalashadi:

$$v_1 = \sqrt{3(1 - 2^{-2/3}) \frac{mnRT}{M(m + M)}}. \quad (5')$$

Porshen idishni tark etganda, idish tezligi qo'shimcha ravishda atomlarning idish tubiga urilishi tufayli v_2 qiymatgacha ortadi. Har bir atom idishga quyidagi impuls beradi:

$$p = 2m_A \Delta \bar{v}_x,$$

bu yerda m_A – atom massasi; $m_A = \frac{M_0}{N_A}$, va $\bar{v}_x$ atomlarning o'rtacha kvadratik tezligi $\bar{v}^2$ orqali topiladi:

$$\bar{v}_x^2 + \bar{v}_y^2 + \bar{v}_z^2 = \bar{v}^2, \quad \bar{v}_x^2 = \bar{v}_y^2 = \bar{v}_z^2, \quad \text{demak } \bar{v}_x = \sqrt{\frac{\bar{v}^2}{3}}.$$

Bir atomning elastik urilishi tufayli idish oladigan o'rtacha impuls:

$$p = 2 \frac{M_0}{N_A} \sqrt{\frac{\bar{v}^2}{3}}.$$

Atomlarning issiqlik tezliklari idish tezligidan ancha katta va harakat idish bilan bog'langan sistemada tasvirlanadi deb faraz qilinadi. Faqat atomlarning yarmi idish tubiga urilishini hisobga olib, idish olgan to'liq impuls:

$$p_t = \frac{1}{2} n N_A p = n M_0 \sqrt{\frac{\bar{v}^2}{3}}. \quad (6)$$

Idish tezligining qo'shimcha ortishi:

$$v_2 = \frac{p_t}{M} = n \frac{M_0}{M} \sqrt{\frac{v^2}{3}}. \quad (7)$$

O'rtacha kvadratik tezlik formulasi $\sqrt{v^2} = \sqrt{\frac{3RT_f}{M_0}}$ va harorat uchun (4) tenglamadan foydalanib, v_2 uchun yakuniy natija:

$$v_2 = 2^{-1/3} \frac{n\sqrt{M_0 RT}}{M}. \quad (8)$$

Demak, idishning oxirgi tezligi:

$$v = v_1 + v_2 = \sqrt{3(1 - 2^{-2/3}) \frac{mnRT}{(nM_0 + M)(m + nM_0 + M)}} + 2^{-1/3} \frac{n\sqrt{M_0 RT}}{M}.$$

Taxminan:

$$v \approx \sqrt{3(1 - 2^{-2/3}) \frac{mnRT}{M(m + M)}} + 2^{-1/3} \frac{n\sqrt{M_0 RT}}{M}. \quad (9)$$

Nazariy masala 2

Qarshiligi $R_0 = 2 \Omega$ bo'lgan, nominal kuchlanishi $U_0 = 4.5 \text{ V}$ da ishlaydigan elektr lampasi EYUK $E = 6 \text{ V}$ va ichki qarshiligi e'tiborga olinmas darajada kichik bo'lgan akkumulyatorga ulangan.

1. Lampaning nominal kuchlanishi, lampa akkumulyatorga qarshiligi R bo'lgan reostat yordamida potensimetrik tarzda ulanganda ta'minlanadi. Agar tizimning foydali ish koeffitsiyenti (FIK) $\eta_0 = 0.6$ dan kichik bo'lmasligi kerak bo'lsa, reostat qarshiligi R qanday bo'lishi va reostatdan oqib o'tadigan maksimal elektr toki $I_{\max}$ qancha bo'lishi kerak?
2. Tizimning mumkin bo'lgan maksimal FIK η qancha va bu holda lampa reostatga qanday ulanishi mumkin?

Nazariy masala 2 yechimi

1. Lampaning U_0 kuchlanishi qarshiligi R bo'lgan reostat yordamida rostlanadi. Kirxgof qonunlaridan foydalanib:

$$I = \frac{U_0}{R} + \frac{U_0}{R - R_x}, \quad (1)$$

$$U_0 = E - IR_x, \quad (2)$$

bu yerda $R - R_x$ – reostatning lampaga parallel ulangan qismining qarshiligi, R_x – qolgan qismning qarshiligi.

Bunday zanjirning FIK η :

$$\eta = \frac{P_{\text{lampa}}}{P_{\text{akkum.}}} = \frac{U_0^2/R_0}{IE} = \frac{U_0^2}{R_0 IE}. \quad (3)$$

(3) tenglamadan ko'rinadiki, reostatdan oqib o'tadigan maksimal tok FIK ning minimal qiymati bilan aniqlanadi:

$$I_{\max} = \frac{U_0^2}{R_0 E \eta_{\min}} = \frac{U_0^2}{R_0 E \eta_0}. \quad (4)$$

Reostat qarshiligi R ning FIK η ga bog'liqligini (3) tenglamadan olingan tok qiymatini, $I = \frac{U_0^2}{R_0 E \eta}$, (1) va (2) tenglamalarga qo'yib aniqlash mumkin:

$$\frac{U_0}{R_0 E \eta} = \frac{1}{R_0} + \frac{1}{R - R_x}, \quad (5)$$

$$R_x = (E - U_0) \frac{R_0 E \eta}{U_0^2}. \quad (6)$$

U holda:

$$R = R_0 \eta \frac{E^2}{U_0^2} \frac{1 + \eta(1 - \frac{E}{U_0})}{1 - \frac{E}{U_0} \eta}. \quad (7)$$

Savollarga javob berish uchun $R(\eta)$ bog'liqlikni tekshirish kerak. Shu sababli birinchi hosila R'_η ni topamiz:

$$R'_\eta \propto \left(\frac{\eta + \eta^2(1 - \frac{E}{U_0})}{1 - \frac{E}{U_0} \eta} \right)' = \eta \left(2 - \frac{E}{U_0} \eta \right) \left(1 - \frac{E}{U_0} \right) + 1.$$

$\eta < 1$ bo'lgani uchun, hosila musbat va $R(\eta)$ funksiya o'suvchi. Demak, reostat qarshiligi minimal bo'lganda FIK minimal bo'ladi. Shuning uchun:

$$R \geq R_{\min} = R_0 \eta_0 \frac{E^2}{U_0^2} \frac{1 + \eta_0(1 - \frac{E}{U_0})}{1 - \frac{E}{U_0} \eta_0} \approx 8.53 \, \Omega.$$

Maksimal tok $I_{\max}$ (4) tenglama yordamida hisoblanadi. Natija: $I_{\max} \approx 660 \text{ mA}$.

2. $R(\eta)$ funksiya o'suvchi bo'lgani uchun, $R \rightarrow \infty$ da $\eta \rightarrow \eta_{\max}$. Bu holda to'liq tok I minimal bo'lib, $\frac{U_0}{R_0}$ ga teng bo'ladi. Demak, maksimal FIK:

$$\eta_{\max} = \frac{U_0}{E} = 0.75.$$

Bu holat reostatni zanjirga uning uchta klemmasidan faqat ikkitasini ishlatib ulash orqali amalga oshiriladi. Reostatning ishlatiladigan qismi R_1 :

$$R_1 = \frac{E - U_0}{I_0} = \frac{E - U_0}{U_0} R_0 \approx 0.67 \, \Omega.$$

Nazariy masala 3

Radioastronomik rasadxonadagi radioto‘lqinlar detektori dengiz sohilida, dengiz sathidan $h = 2$ m balandlikda joylashtirilgan. Gorizont ustida $\lambda = 21$ sm to‘lqin uzunlikdagi elektromagnit to‘lqinlar chiqaradigan yulduz ko‘tarilgach, detektor ketma-ket maksimum va minimumlarni qayd etadi. Qayd etilgan signal qabul qilingan to‘lqinlar intensivligiga proporsional. Detektor elektr vektori dengiz sirtiga parallel yo‘nalishda tebranadigan to‘lqinlarni qayd etadi.

1. Detektor maksimum va minimumlarni qayd etadigan paytdagi yulduz va gorizont orasidagi burchakni (umumiy holda) aniqlang.
2. Yulduz ko‘tarilgandan so‘ng darhol signal kamayadimi yoki ortadimi?
3. Birinchi maksimumning keyingi minimumga nisbatan signal nisbatini aniqlang. Elektromagnit to‘lqinning suv sirtidan qaytishida, qaytgan (E_r) va tushayotgan (E_i) to‘lqinlar elektr maydoni intensivliklarining nisbati quyidagi qonunga bo‘ysunadi:

$$\frac{E_r}{E_i} = \frac{n - \cos \phi}{n + \cos \phi},$$

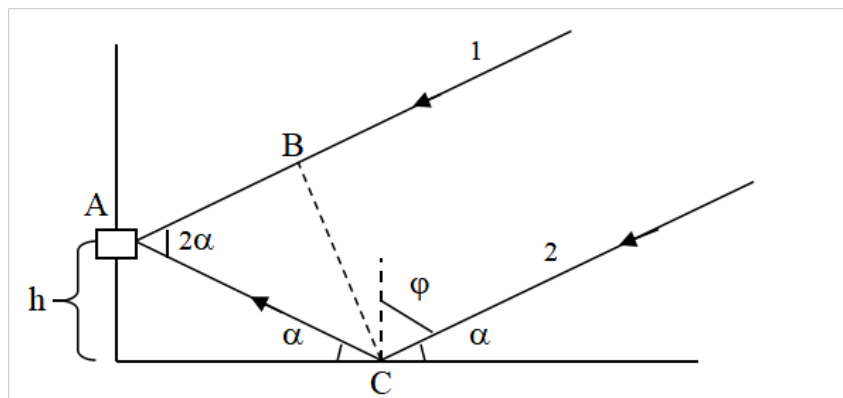
bu yerda n – sindirish ko‘rsatkichi, ϕ – to‘lqinning tushish burchagi. “Havo-suv” sirti uchun $\lambda = 21$ sm da sindirish ko‘rsatkichi $n = 9$.

4. Yulduzning ko‘tarilishi bilan ketma-ket maksimum va minimumlar intensivliklarining nisbati ortadimi yoki kamayadimi?

Dengiz sirti tekis deb faraz qilinsin.

Nazariy masala 3 yechimi

1. Detektor A qayd etgan signal ikki nurning interferensiyasi natijasidir: to‘g‘ridan-to‘g‘ri yulduzdan kelayotgan 1-nur va dengiz sirtidan qaytgan 2-nur (rasmga qarang).



Interferensiya sxemasi

Ikkinchi nurning fazasi kattaroq sindirish ko'rsatkichli muhitdan qaytish tufayli π ga siljiydi. Shuning uchun ikki nur orasidagi fazalar farqi:

$$\begin{aligned}\Delta &= AC + \frac{\lambda}{2} - AB = \frac{h}{\sin \alpha} + \frac{\lambda}{2} - \left(\frac{h}{\sin \alpha} \right) \cos(2\alpha) \\ &= \frac{\lambda}{2} + \frac{h}{\sin \alpha} [1 - \cos(2\alpha)] = \frac{\lambda}{2} + 2h \sin \alpha.\end{aligned}\quad (1)$$

Interferension maksimum sharti:

$$\frac{\lambda}{2} + 2h \sin \alpha_{\max} = k\lambda, \quad \text{yoki} \quad \sin \alpha_{\max} = \left(k - \frac{1}{2}\right) \frac{\lambda}{2h} = (2k - 1) \frac{\lambda}{4h}, \quad (2)$$

bu yerda $k = 1, 2, 3, \dots, 19$. (optik yo'llar farqi $2h$ dan oshmasligi sababli k 19 dan katta bo'la olmaydi).

Interferension minimum sharti:

$$\frac{\lambda}{2} + 2h \sin \alpha_{\min} = (2k + 1) \frac{\lambda}{2}, \quad \text{yoki} \quad \sin \alpha_{\min} = \frac{k\lambda}{2h}, \quad (3)$$

bu yerda $k = 1, 2, 3, \dots, 19$.

2. Yulduz ko'tarilgandan so'ng darhol burchak balandligi α nolga teng, shuning uchun interferension minimum sharti bajariladi. Shu sababli, yulduz ko'tarilgandan so'ng darhol signal ortadi.

3. Agar interferension maksimum sharti bajarilsa, elektr maydon intensivligi to'g'ri nur E_i va qaytgan nur E_r intensivliklarining yig'indisiga teng: $E_{\max} = E_i + E_r$. $E_r = E_i \frac{n - \cos \phi}{n + \cos \phi}$ ekanligidan, $E_{\max} = E_i \left(1 + \frac{n - \cos \phi_{\max}}{n + \cos \phi_{\max}}\right)$. Rasmdan $\phi_{\max} = \frac{\pi}{2} - \alpha_{\max}$ ekanligi ko'rinadi, demak:

$$E_{\max} = E_i \left(1 + \frac{n - \sin \alpha_{\max}}{n + \sin \alpha_{\max}}\right) = E_i \frac{2n}{n + \sin(2\alpha_{\max})}. \quad (4)$$

Interferension minimumda natijaviy intensivlik:

$$E_{\min} = E_i - E_r = E_i \frac{2 \sin \alpha_{\min}}{n + \sin \alpha_{\min}}. \quad (5)$$

Signal intensivligi I elektr maydon intensivligi E kvadratiga proporsional, shuning uchun ketma-ket maksimum va minimumlar intensivliklarining nisbati:

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{E_{\max}}{E_{\min}} \right)^2 = \frac{n^2}{\sin^2 \alpha_{\min}} \frac{(n + \sin \alpha_{\min})^2}{(n + \sin \alpha_{\max})^2}. \quad (6)$$

(2) va (3) tenglamalardan foydalanib, (6) tenglamani quyidagi ko‘rinishga keltirish mumkin:

$$\frac{I_{\max}}{I_{\min}} = \frac{4n^2h^2}{k^2\lambda^2} \left[\frac{n + k\frac{\lambda}{2h}}{n + (2k-1)\frac{\lambda}{4h}} \right]^2. \quad (7)$$

Ushbu umumiy formuladan birinchi maksimum ($k = 1$) va keyingi minimum uchun nisbatni aniqlaymiz:

$$\frac{I_{\max}}{I_{\min}} = \frac{4n^2h^2}{\lambda^2} \left(\frac{n + \frac{\lambda}{2h}}{n + \frac{\lambda}{4h}} \right)^2 = 3 \cdot 10^4.$$

4. $n \gg \frac{\lambda}{2h}$ ekanligidan, (7) tenglamadan quyidagi kelib chiqadi:

$$\frac{I_{\max}}{I_{\min}} \approx \frac{4n^2h^2}{k^2\lambda^2}.$$

Shunday qilib, yulduzning ko‘tarilishi bilan ketma-ket maksimum va minimumlar intensivliklarining nisbati kamayadi.

Eksperimental masala

Materiallar va asboblari: elastik rezina shnur (erkin shnur uzunligi $l_0 = 150$ mm), shtativga vertikal osilgan, 10 g dan 100 g gacha bo'lgan toshlar to'plami, massasi 5 g bo'lgan toshlar uchun patnis, sekundomer, chizg'ich, millimetrli qog'oz.

Eslatma: Yerning tortishish tezlanishi $g = 10$ m/s². Rezina shnurning massasini hisobga olmasa bo'ladi.

Quyidagi tadqiqotni bajaring:

1. Rezina shnurni 15 g dan 105 g gacha bo'lgan oraliqdagi toshlar bilan yuklang. Olingan ma'lumotlarni jadvalga kiriting. Shnurning cho'zilishining kuchlanish kuchi F ga bog'liqligini eksperimental ravishda olingan grafikda (mos masshtabda) tasvirlang.
2. 1-bandda olingan eksperimental natijalardan foydalanib, shnur hajmini 35 g dan 95 g gacha bo'lgan yuklanish oralig'ida hisoblang va jadvalga kiriting. Hisoblashlarni ushbu oraliqdagi har bir ikkita qo'shni yuklanish qiymati uchun ketma-ket bajaring. Hisoblashlar uchun foydalangan formulalaringizni yozing. Hajmning yuklanishga bog'liqligi haqida analitik taklif kiriting.

Yung moduli o'zgaras deb faraz qilinsin: $E = 2 \cdot 10^6$ Pa. Shuni yodda tutingki, Guk qonuni faqat taxminan bajariladi va undan chetlanishlar 10% gacha bo'lishi mumkin.

3. Og'irlik massasi 60 g bo'lganda, sekundomer yordamida rezina shnurning hajmini aniqlang. Foydalanilgan formulalarni yozing.

Eksperimental masala yechimi

1. Har xil yuklanish m_n da shnur uzunligi l_n kamida 10 marta o'lchanishi kerak. Natijalar 1-jadvalda keltirilgan.

Table 1: 1-jadval

m_n , kg	$F_n = m_n g$, N	l_n , mm	$\Delta l_n = l_n - l_0$, mm
0.005	0.05	153	3
0.015	0.15	158	8
0.025	0.25	164	14
0.035	0.35	172	22
0.045	0.45	181	31
0.055	0.55	191	41
0.065	0.65	202	53
0.075	0.75	215	65
0.085	0.85	228	78
0.095	0.95	243	93
0.105	1.05	261	111

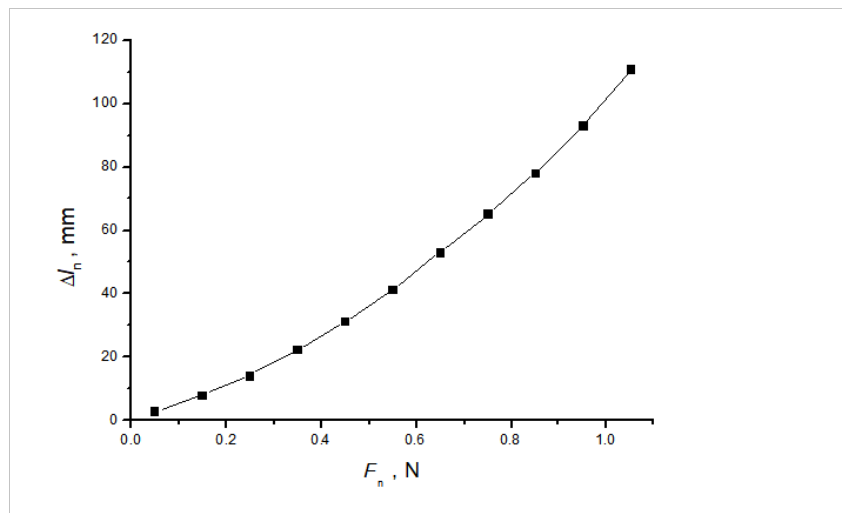


Figure 1: 1-rasm: Cho‘zilishning kuchga bog‘liqligi

Shnur cho‘zilishining kuchlanish kuchi F ga bog‘liqligi grafigi 1-rasmda ko‘rsatilgan.

2. Hajmni hisoblash uchun har bir o‘lchash uchun Guk qonunidan foydalanish mumkin:

$$\frac{\Delta l'_n}{l_n} = \frac{1}{E} \frac{\Delta F_n}{S_n},$$

demak,

$$S_n = \frac{l_n \Delta F_n}{E \Delta l'_n},$$

bu yerda $\Delta l'_n = l_n - l_{n-1}$, $\Delta F_n = \Delta m \cdot g$. ($\frac{\Delta l_n}{l_n} = \frac{1}{E} \frac{F_n}{S_n}$ ko‘rinishidagi Guk qonunidan foydalanish kattaroq xatolikka olib keladi, chunki Δl_n ning qiymati l_n bilan bir xil tartibda).

S_n ning qiymati aniqlangach, har bir F_n qiymatida hajm V_n ni hisoblash oson:

$$V_n = S_n l_n = \frac{l_n^2 \Delta F_n}{E \Delta l'_n}.$$

1-jadval ma’lumotlaridan foydalanib, barcha hisoblashlar 2-jadvalda keltirilgan:

Table 2: 2-jadval

$\Delta m_n, \text{kg}$	$\Delta F_n, \text{N}$	l_n, m	$\Delta l'_n, \text{m}$	S_n, m^2	V_n, m^3
0.035–0.025	0.1	0.172	0.008	$1.07 \cdot 10^{-6}$	$184 \cdot 10^{-9}$
0.045–0.035	0.1	0.181	0.009	$1.01 \cdot 10^{-6}$	$183 \cdot 10^{-9}$
0.055–0.045	0.1	0.191	0.010	$0.95 \cdot 10^{-6}$	$182 \cdot 10^{-9}$
0.065–0.055	0.1	0.203	0.012	$0.92 \cdot 10^{-6}$	$187 \cdot 10^{-9}$
0.075–0.065	0.1	0.215	0.012	$0.89 \cdot 10^{-6}$	$191 \cdot 10^{-9}$
0.085–0.075	0.1	0.228	0.013	$0.88 \cdot 10^{-6}$	$200 \cdot 10^{-9}$
0.095–0.085	0.1	0.243	0.015	$0.81 \cdot 10^{-6}$	$196 \cdot 10^{-9}$
0.105–0.095	0.1	0.261	0.018	$0.72 \cdot 10^{-6}$	$188 \cdot 10^{-9}$

Natijalar shuni ko‘rsatadiki, hajmning hisoblangan qiymatlarining o‘rtacha qiymatdan

nisbiy chetlanishi:

$$\varepsilon = \frac{\Delta V_{n,o'rt.} \cdot 100\%}{V_{o'rt.}} \approx \frac{5.3 \cdot 10^{-9}}{189 \cdot 10^{-9}} \cdot 100\% \approx 2.8\%.$$

Shuning uchun xulosa shuki, rezina shnurning cho'zilishdagi hajmi o'zgarmasdir: $V_n = \text{const}$.

3. Qayd etilgan yuklanishda rezina shnurning hajmini shnurning kichik tebranishlarini tekshirish orqali aniqlash mumkin. Bu tebranishlarning sababi elastik kuchdir:

$$F = ES \frac{\Delta l}{l}.$$

Nyutonning ikkinchi qonunidan foydalanib:

$$-ES \frac{\Delta l}{l} = m \frac{d^2(\Delta l)}{dt^2},$$

tebranishlar davri aniqlanishi mumkin:

$$T = 2\pi \sqrt{\frac{ml}{ES}}.$$

U holda:

$$S = \frac{(2\pi)^2 ml}{ET^2},$$

va shnurning hajmi:

$$V = Sl = \frac{4\pi^2 ml^2}{ET^2}.$$

Davrni o'lchash quyidagini beradi: $T = t/n = 5.25 \text{ s}/10 = 0.52 \text{ s}$, bunda foydalanilgan massa $m = 0.065 \text{ kg}$. Hajm natijasi $V \approx 195 \cdot 10^{-9} \text{ m}^3$, bu 2-qismda olingan natijalarga mos keladi.